

EXPERIMENT – 12

Title

To Implement Cursor on a Table

Aim / Objective

To understand and implement **PL/SQL Cursors** for retrieving and processing records from a database table one row at a time.

Theory

A **Cursor** is a pointer to the result set of an SQL query. It allows PL/SQL to process query results row by row instead of all at once.

Oracle automatically creates an **Implicit Cursor** for SQL statements such as **INSERT, UPDATE, DELETE, and SELECT INTO**. When multiple rows are returned by a query, a programmer must use an **Explicit Cursor** to process each row individually.

A cursor follows four basic steps:

1. Declare the Cursor
 2. Open the Cursor
 3. Fetch Records
 4. Close the Cursor
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Types of Cursors

1. Implicit Cursor

- Automatically created by Oracle.
 - Used for DML statements and SELECT INTO.
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2. Explicit Cursor

- Declared by the programmer.
 - Used when a query returns multiple rows.
 - Requires OPEN, FETCH, and CLOSE statements.
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Cursor Attributes

Oracle provides the following cursor attributes:

Attribute	Description
%FOUND	Returns TRUE if a row is fetched successfully
%NOTFOUND	Returns TRUE when no more rows are available
%ROWCOUNT	Returns the number of rows processed
%ISOPEN	Returns TRUE if the cursor is open

Learning Objectives

After completing this experiment, students will be able to:

- Understand implicit and explicit cursors.
 - Declare and use explicit cursors.
 - Fetch records one by one.
 - Display records using DBMS_OUTPUT.
 - Use cursor attributes effectively.
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Software Requirements

- Oracle Database
 - Oracle SQL Developer / SQL*Plus
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Algorithm / Procedure

1. Open Oracle SQL Developer.
2. Connect to the Oracle database.
3. Create the Employee table.
4. Insert sample records.
5. Declare an explicit cursor.
6. Open the cursor.
7. Fetch records.

8. Display the records.
 9. Close the cursor.
 10. Stop.
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Program

Step 1 : Create Employee Table

```
CREATE TABLE Employee
(
EmpID NUMBER PRIMARY KEY,
EmpName VARCHAR2(30),
Department VARCHAR2(30),
Salary NUMBER(10,2)
);
```

Step 2 : Insert Sample Records

```
INSERT INTO Employee VALUES(101,'Rahul','IT',35000);

INSERT INTO Employee VALUES(102,'Priya','HR',42000);

INSERT INTO Employee VALUES(103,'Aman','Finance',39000);

INSERT INTO Employee VALUES(104,'Sneha','Marketing',45000);

COMMIT;
```

Program 1 : Explicit Cursor

```
DECLARE

CURSOR Emp_Cursor IS

SELECT EmpID,EmpName,Salary

FROM Employee;

V_ID Employee.EmpID%TYPE;

V_Name Employee.EmpName%TYPE;

V_Salary Employee.Salary%TYPE;

BEGIN

OPEN Emp_Cursor;
```

```
LOOP

FETCH Emp_Cursor INTO V_ID,V_Name,V_Salary;

EXIT WHEN Emp_Cursor%NOTFOUND;

DBMS_OUTPUT.PUT_LINE
(
V_ID||' '||
V_Name||' '||
V_Salary
);

END LOOP;

CLOSE Emp_Cursor;

END;
/
```

Expected Output

101 Rahul 35000

102 Priya 42000

103 Aman 39000

104 Sneha 45000

PL/SQL procedure successfully completed.

Program 2 : Cursor with WHERE Condition

```
DECLARE

CURSOR IT_Cursor IS

SELECT EmpName,Salary

FROM Employee

WHERE Department='IT';

V_Name Employee.EmpName%TYPE;

V_Salary Employee.Salary%TYPE;

BEGIN
```

```
OPEN IT_Cursor;

LOOP

FETCH IT_Cursor INTO V_Name,V_Salary;

EXIT WHEN IT_Cursor%NOTFOUND;

DBMS_OUTPUT.PUT_LINE
(
V_Name||' '||V_Salary
);

END LOOP;

CLOSE IT_Cursor;

END;
/
```

Expected Output

Rahul 35000

Program 3 : Cursor FOR LOOP

```
BEGIN

FOR Emp_Record IN

(
SELECT *

FROM Employee
)

LOOP

DBMS_OUTPUT.PUT_LINE
(
Emp_Record.EmpID||' '||
Emp_Record.EmpName||' '||
Emp_Record.Department
);

END LOOP;

END;
/
```

Expected Output

101 Rahul IT

102 Priya HR

103 Aman Finance

104 Sneha Marketing

Program 4 : Cursor Using %ROWCOUNT

DECLARE

CURSOR C1 IS

SELECT *

FROM Employee;

V_Record Employee%ROWTYPE;

BEGIN

OPEN C1;

LOOP

FETCH C1 INTO V_Record;

EXIT WHEN C1%NOTFOUND;

DBMS_OUTPUT.PUT_LINE

(
'Row Number : '||
C1%ROWCOUNT||
' Employee : '||
V_Record.EmpName
);

END LOOP;

CLOSE C1;

END;

/

Expected Output

Row Number : 1 Employee : Rahul

Row Number : 2 Employee : Priya

Row Number : 3 Employee : Aman

Row Number : 4 Employee : Sneha

Program 5 : Implicit Cursor

BEGIN

UPDATE Employee

SET Salary=Salary+2000

WHERE Department='IT';

DBMS_OUTPUT.PUT_LINE

(
'Rows Updated : '||
SQL%ROWCOUNT
);

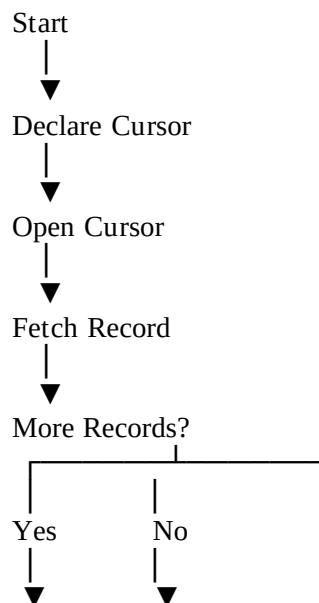
END;

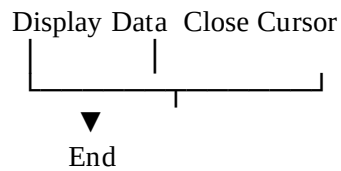
/

Expected Output

Rows Updated : 1

Flowchart





Explanation

Cursor

A cursor is used to retrieve multiple rows one at a time.

OPEN

Allocates memory and executes the query.

FETCH

Retrieves one row from the cursor.

CLOSE

Releases the cursor resources.

Cursor FOR LOOP

Automatically opens, fetches, and closes the cursor.

Cursor Attributes

- **%FOUND** → Indicates whether a row was fetched.
 - **%NOTFOUND** → Indicates no more rows are available.
 - **%ROWCOUNT** → Returns the number of rows processed.
 - **%ISOPEN** → Indicates whether the cursor is currently open.
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Observation

- Explicit cursors processed multiple records successfully.

- Cursor FOR LOOP simplified cursor handling.
 - Cursor attributes returned accurate information.
 - Implicit cursor displayed the number of rows updated.
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Result

PL/SQL cursors were successfully implemented to retrieve and process records from the Employee table. Both implicit and explicit cursors were executed successfully, demonstrating efficient row-by-row processing in Oracle PL/SQL.

Applications

- Payroll Processing
 - Banking Systems
 - Employee Management Systems
 - Hospital Management Systems
 - Student Information Systems
 - Inventory Management Systems
 - Financial Reporting
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Advantages

- Processes multiple records efficiently.
 - Supports row-by-row data processing.
 - Provides better control over query results.
 - Useful for complex business logic.
 - Supports cursor attributes for monitoring processing.
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Viva Questions

1. What is a cursor in PL/SQL?
2. What is the difference between an implicit and an explicit cursor?
3. Why is an explicit cursor required?

4. What are the four steps involved in using an explicit cursor?
 5. What is the purpose of the OPEN statement?
 6. What does the FETCH statement do?
 7. Why is the CLOSE statement required?
 8. What is a Cursor FOR LOOP?
 9. Explain the cursor attributes %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN.
 10. What is the difference between %ROWTYPE and %TYPE?
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Lab Assignment

1. Create a cursor to display all employee details.
 2. Display employees with a salary greater than ₹40,000 using a cursor.
 3. Use a cursor to calculate the total salary of all employees.
 4. Display department-wise employee details using an explicit cursor.
 5. Demonstrate the use of %FOUND and %NOTFOUND.
 6. Write a PL/SQL program using a Cursor FOR LOOP.
 7. Update employee salaries and display the number of rows affected using SQL%ROWCOUNT.
 8. Compare implicit and explicit cursors with suitable examples.
 9. Write a cursor program using %ROWTYPE.
 10. Explain the applications of cursors in database programming.
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Precautions

- Always close an explicit cursor after processing records.
 - Use EXIT WHEN cursor_name%NOTFOUND to avoid infinite loops.
 - Enable **DBMS Output** before executing PL/SQL programs.
 - Use %TYPE and %ROWTYPE to ensure compatibility with table column data types.
 - Avoid unnecessary cursors for simple SQL operations, as set-based SQL is generally more efficient.
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Learning Outcome

After completing this experiment, students will be able to:

- Understand the concept and purpose of cursors in PL/SQL.
- Differentiate between implicit and explicit cursors.
- Implement explicit cursors to process multiple rows.
- Use cursor attributes to monitor cursor operations.
- Apply cursors in real-world database applications requiring row-by-row processing.